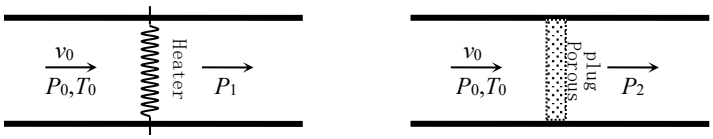
Ideal gas, molar mass μ, flows right in a long straight smooth horizontal adiabatic pipe of area S. Molar internal energy at absolute T is $\frac{5}{2}RT$; R is the gas constant.

（1） A fixed mid-pipe heater supplies constant power W, Fig. （a）. Neglect its resistance to flow. In steady flow the state is nonuniform near the heater, becoming uniform far away. In the uniform steady region on the left, pressure is p_0, temperature T_0, and rightward speed v_0. On the right, steady pressure is p_1. Find that region's temperature T_1.

（2） Use a porous plug instead, Fig. （b）. Left steady temperature and pressure are T_0, p_0, with speed v_0 rightward and right pressure p_2（$p_2<p_0$）.

With no energy exchange of any kind between gas and plug, find the downstream flow speed v_2.



（a） （b）

Fig. 28 Final—4

V. （15 points）

Prism ABC has apex angle α below 90°. A ray in the paper plane hits AB at D at any angle. After one refraction it hits AC, It can reflect totally at AC. The critical angle at AC is θ（$\theta<45°$）, and AC is long enough. Find the range of prism apex α for the following two cases:

（1） The ray enters AB only below the normal;

（2） The ray enters AB only above the normal.

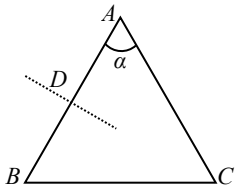


Fig. 28 Final—5

VI. （20 points）

Charge q gives an electric field at distance r of magnitude $E=k_e\frac{q}{r^2}$, with constant k_e. A long straight wire carrying i produces a field at distance r（far below its length ）, of magnitude $B=k_m\frac{2i}{r}$, with constant k_m.

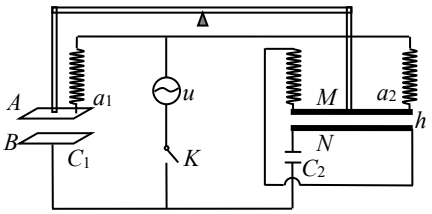


Fig. 28 Final—6

Measure $\sqrt{\frac{k_e}{k_m}}$ in the shown setup using low-frequency （hundreds of hertz） electric and magnetic fields. A、B are horizontal square plates of capacitance C_1. Side a_1（gap far below a_1）. B is fixed; A hangs on one balance hook.

M、N are parallel horizontal thin metal rods: length a_2, gap h（$h\ll a_2$）. Rod N is fixed; M hangs on the other hook. C_2 is a known capacitance; K is a switch. AC voltage u varies with time t as $u=U_0\cos 2\pi ft$; f is frequency. Wire all parts as shown. The balance is level with K open, but loses balance when powered. By